MSG No. 139.536 $I4'/m'm'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m'm'm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m'm'm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'm'm'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8,9,10,11,12]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,18,19,20,29,30,31,32]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[21,22,23,24,25,26,27,28]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,14,15,21,22,27,28]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,13,16,23,24,25,26]`
3	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[5,6,11,12,17,20,30,31]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,8,9,10,18,19,29,32]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'm'm`

No.	position	mapping
1	$[0, 0, z]$	`[1,4,7,8,9,10,14,15]`
2	$[0, 0, -z]$	`[2,3,5,6,11,12,13,16]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[17,20,23,24,25,26,30,31]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[18,19,21,22,27,28,29,32]`

Table 6: Wyckoff site: `8f`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,8,28,29]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,5,25,30]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,6,26,31]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,7,27,32]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9,14,18,21]
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[10,15,19,22]
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[11,16,20,23]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[12,13,17,24]

Table 7: Wyckoff site: 8g, site symmetry: 2m'm'.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,4,14,15]
2	$\left[0, \frac{1}{2}, -z\right]$	[2,3,13,16]
3	$\left[\frac{1}{2}, 0, -z\right]$	[5,6,11,12]
4	$\left[\frac{1}{2}, 0, z\right]$	[7,8,9,10]
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[17,20,30,31]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[18,19,29,32]
7	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[21,22,27,28]
8	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[23,24,25,26]

Table 8: Wyckoff site: 8h, site symmetry: m'.2'm

No.	position	mapping
1	$[x, x, 0]$	[1,8,11,16]
2	$[x, -x, 0]$	[2,5,10,15]
3	$[-x, x, 0]$	[3,6,9,14]
4	$[-x, -x, 0]$	[4,7,12,13]
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[17,24,27,32]
6	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[18,21,26,31]
7	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	[19,22,25,30]
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	[20,23,28,29]

Table 9: Wyckoff site: 8i, site symmetry: m'2m'.

No.	position	mapping
1	$[x, 0, 0]$	[1,2,15,16]
2	$[-x, 0, 0]$	[3,4,13,14]
3	$[0, -x, 0]$	[5,7,10,12]
4	$[0, x, 0]$	[6,8,9,11]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,18,31,32]
6	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[19,20,29,30]
7	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[21,23,26,28]
8	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[22,24,25,27]

Table 10: Wyckoff site: 8j, site symmetry: m'2m'.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,2,15,16]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,4,13,14]
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	[5,7,10,12]
4	$\left[\frac{1}{2},\ x,\ 0\right]$	[6,8,9,11]
5	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[17,18,31,32]
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[19,20,29,30]
7	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[21,23,26,28]
8	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[22,24,25,27]

Table 11: Wyckoff site: 16k, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,27]
2	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[2,26]
3	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[3,25]
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[4,28]
5	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[5,31]
6	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[6,30]
7	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[7,29]
8	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[8,32]
9	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[9,19]
10	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[10,18]
11	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[11,17]
12	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[12,20]
13	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[13,23]
14	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[14,22]
15	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[15,21]
16	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[16,24]

Table 12: Wyckoff site: 16l, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,16]
2	$[x,\ -y,\ 0]$	[2,15]
3	$[-x,\ y,\ 0]$	[3,14]
4	$[-x,\ -y,\ 0]$	[4,13]
5	$[y,\ -x,\ 0]$	[5,10]
6	$[-y,\ x,\ 0]$	[6,9]
7	$[-y,\ -x,\ 0]$	[7,12]
8	$[y,\ x,\ 0]$	[8,11]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,32]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[18,31]
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[19,30]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[20,29]

continued ...

Table 12

No.	position	mapping
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[21,26]
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[22,25]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[23,28]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[24,27]

Table 13: Wyckoff site: 16m, site symmetry: ..m

No.	position	mapping
1	$[x,\ x,\ z]$	[1,8]
2	$[x,\ -x,\ -z]$	[2,5]
3	$[-x,\ x,\ -z]$	[3,6]
4	$[-x,\ -x,\ z]$	[4,7]
5	$[-x,\ x,\ z]$	[9,14]
6	$[x,\ -x,\ z]$	[10,15]
7	$[x,\ x,\ -z]$	[11,16]
8	$[-x,\ -x,\ -z]$	[12,13]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,24]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[18,21]
11	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19,22]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[20,23]
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25,30]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[26,31]
15	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27,32]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[28,29]

Table 14: Wyckoff site: 16n, site symmetry: .m'.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,14]
2	$[0,\ -y,\ -z]$	[2,13]
3	$[0,\ y,\ -z]$	[3,16]
4	$[0,\ -y,\ z]$	[4,15]
5	$[y,\ 0,\ -z]$	[5,11]
6	$[-y,\ 0,\ -z]$	[6,12]
7	$[-y,\ 0,\ z]$	[7,9]
8	$[y,\ 0,\ z]$	[8,10]
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,30]
10	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[18,29]
11	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19,32]
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20,31]
13	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[21,27]
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[22,28]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[23,25]
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[24,26]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$[-y,\ -x,\ z]$	[7]
8	$[y,\ x,\ z]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]
11	$[y,\ x,\ -z]$	[11]
12	$[-y,\ -x,\ -z]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[-x,\ y,\ z]$	[14]
15	$[x,\ -y,\ z]$	[15]
16	$[x,\ y,\ -z]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[18]
19	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19]
20	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[23]
24	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[24]
25	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25]
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[26]
27	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[30]
31	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[31]
32	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[32]